

Artificial Intelligence Assignment 1

Performance Evaluation Report

1 Introduction

We are building a model to detect the digit being spoken when it is fed with a audio file , our range is 0 to 9 , we have 6 speakers and over 3000 voice recordings to build a model, we are required to try out Both HMM and GMM models and then we have to decide which is better one for this setup.

This report evaluates the performance of two distinct modeling approaches :

- * Gaussian Mixture Model (GMM)
- * discrete Hidden Markov Model(HMM) with a 128-size codebook.

The models were trained on audio files from Speakers 1–5 and evaluated on a validation set (80/20 split) and test on the recordings of speaker-6

2 System Architecture & Algorithms

By feature extraction first we convert the audio files into compact set of numbers, that represent the "signature" of the sound, for this I used "Mel-Frequency Cepstral Coefficients (MFCCs)" as it is what best used in the industry. It is Recommended. Captures the unique resonance of human speech

- **Continuous GMM:** In the training if model I used "5" Gaussian components per digit. To ensure convergence to a global optimum, and the I used K-means initilisation for cluster centers and Used full covariance matrices even though it is computationally heavier for accuracy . The Log-Sum-Exp was implemented to maintain numerical stability during the (EM) loop .This is basic idea in the GMM I used.
- **Discrete HMM:** Utilized Vector Quantization (VQ) with a codebook size of 128 clusters. The temporal structure was modeled using a Left-to-Right topology, and with parameters estimated via the Baum-Welch algorithm.
- **Preprocessing:** As the audio files contains empty noises, I trimmed down the empty noises to improve the accuracy of both models.

For each digit I trained both GMM and HMM models , and in the final prediction we run the audio files for all 10 models for Both GMM and HMM and then we select the one with maximum probability as the predicted digit, for each GMM and HMM. This is how This architecture detects the digit.

3 Overall Metric Comparison

The GMM model significantly outperformed the HMM model across both validation and generalization metrics. but both models face the problem of limited training data set in this case to aim for higher accuracies, As sir has mentioned to train the models on speakers-1-5 , I split the total recordings of first 5 speakers into 80 percent train set and 20 percent validation set randomly to ensure better generality

Table 1: Comparative Overall Accuracy

Metric	GMM Accuracy	HMM Accuracy	Performance Gap
Validation (80/20)	97.20%	87.20%	10.00%
Test (Speaker 6)	81.00%	67.00%	14.00%

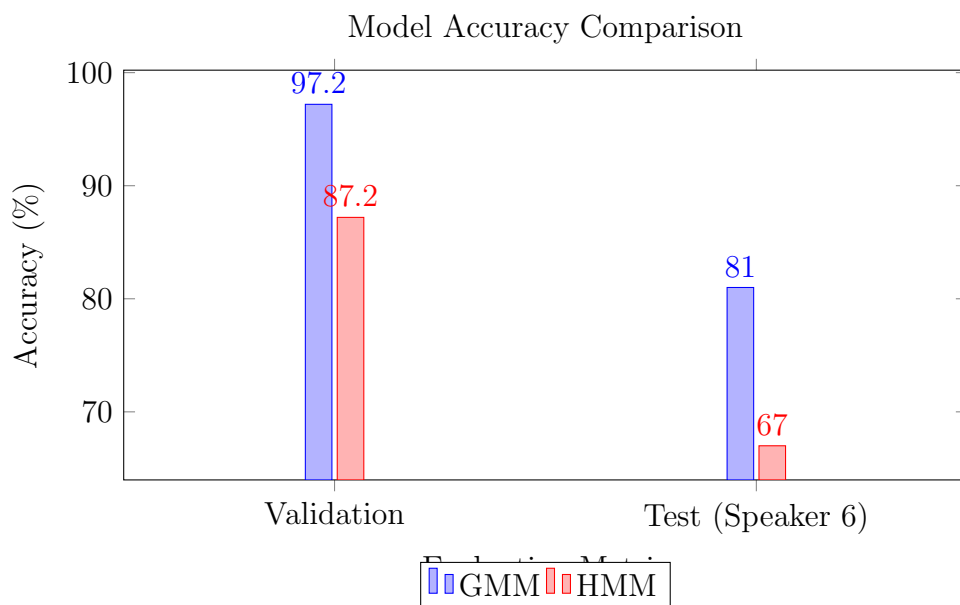


Figure 1: Comparison of overall accuracy between Continuous GMM and Discrete HMM. The y-axis represents the percentage accuracy, and the x-axis represents the dataset split.

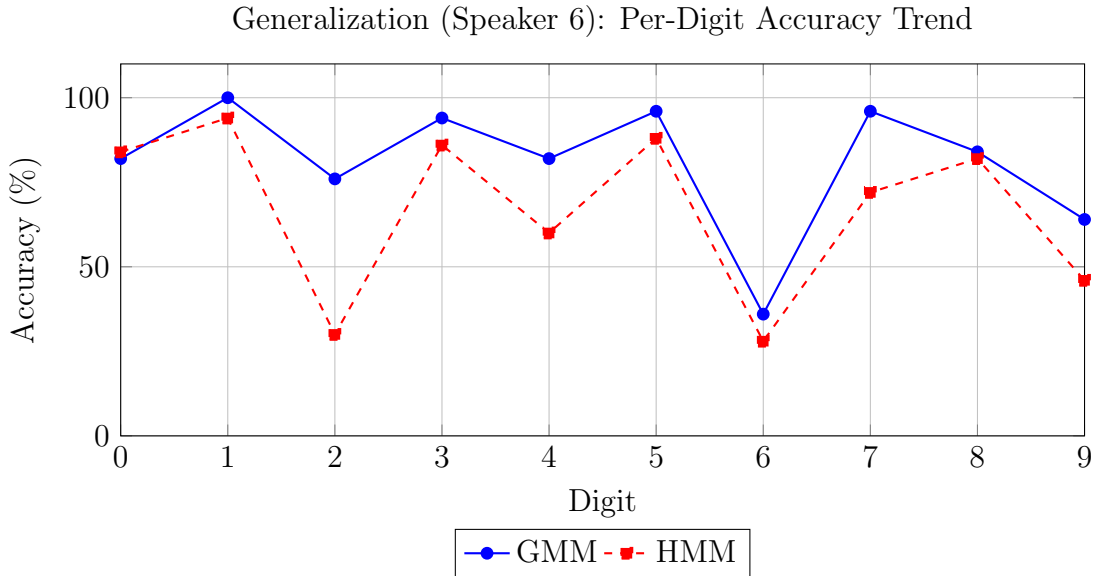
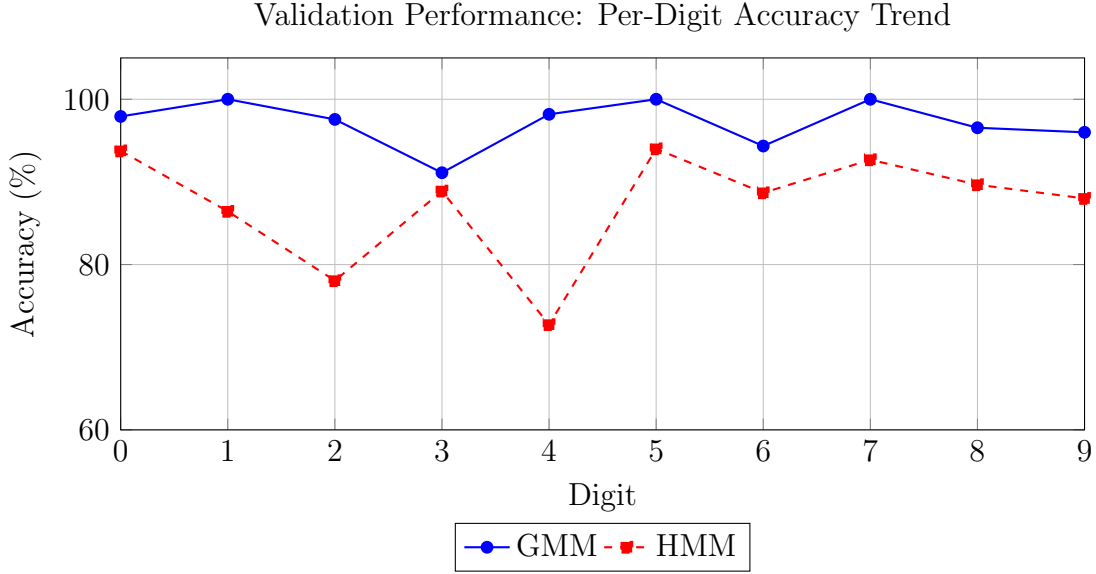
4 Per-Digit Performance Analysis

The performance of the Continuous GMM and Discrete HMM models varies significantly based on the phonetic complexity of each digit. The following table summarizes the test results for Speaker 6.

The per-digit breakdown reveals that while both models handle digits like “1” and “5” with high precision, the GMM demonstrates superior robustness on challenging vowels.

Table 2: Per-Digit Accuracy Breakdown (Test Set - Speaker 6)

Model	0	1	2	3	4	5	6	7	8	9
GMM (%)	82.0	100.0	76.0	94.0	82.0	96.0	36.0	96.0	84.0	64.0
HMM (%)	84.0	94.0	30.0	86.0	60.0	88.0	28.0	72.0	82.0	46.0



4.1 Key Insights from per-digit analysis

Clear Vowel Sounds (Digits 1, 2, 5, 7): The GMM is much better for these digits because it can follow the exact shape of a new voice. The HMM fails here because it tries to force sounds into groups, which caused it to mistake "Two" for "Zero." The GMM stays accurate because its math is flexible enough to handle a speaker it has never heard before.

The "S" Sound Problem (Digit 6): Neither model is particularly good for this digit, as it caused the most mistakes for both. The "S" sound is just high-pitched noise that looks too much like background static or the breathy start of the word "Three." Because the models cannot tell these noisy patterns apart, they often guess the wrong digit.

Similar Endings (Digits 1, 9): The GMM is the better choice for these digits because it looks at the entire sound as a single "fingerprint." The HMM tries to follow the word bit-by-bit and often gets confused when both words end with the same "N" sound. By looking at the whole word at once, the GMM avoids swapping "One" and "Nine."

The **Continuous GMM** is the more appropriate model for isolated digits. It is significantly more robust to speaker variation because it maintains the integrity of the MFCC features, while the HMM's discrete codebook introduces a bottleneck that collapses phonetic distinctions.

5 Confusion Matrices Data

Table 3: Complete Confusion Matrix Block

(a) GMM Validation										
	0	1	2	3	4	5	6	7	8	9
0	47	0	1	0	0	0	0	0	0	0
1	0	59	0	0	0	0	0	0	0	0
2	0	0	40	0	0	0	1	0	0	0
3	0	0	2	41	0	0	2	0	0	0
4	0	1	0	0	54	0	0	0	0	0
5	0	0	0	0	0	50	0	0	0	0
6	0	0	1	0	0	0	50	1	1	0
7	0	0	0	0	0	0	0	41	0	0
8	0	0	0	0	0	0	1	1	56	0
9	0	0	0	0	0	0	0	2	0	48

(b) HMM Validation										
	0	1	2	3	4	5	6	7	8	9
0	45	0	0	0	1	0	1	1	0	0
1	0	51	0	0	2	3	0	2	0	1
2	1	0	32	1	5	0	1	0	0	1
3	0	0	0	40	1	0	2	2	0	0
4	0	4	2	0	40	1	0	6	0	2
5	0	1	0	0	1	47	0	1	0	0
6	0	1	1	0	0	1	47	0	3	0
7	0	0	0	0	1	0	0	38	0	2
8	0	0	0	0	0	0	3	3	52	0
9	0	0	0	0	0	1	0	5	0	44

(c) GMM Test (Speaker 6)										
	0	1	2	3	4	5	6	7	8	9
0	41	0	6	0	0	0	1	2	0	0
1	0	50	0	0	0	0	0	0	0	0
2	3	0	38	0	3	0	1	3	2	0
3	0	0	0	47	0	0	0	1	2	0
4	0	2	1	0	41	0	2	2	2	0
5	0	1	0	0	0	48	0	1	0	0
6	0	1	1	18	0	0	18	3	8	1
7	0	0	0	0	0	0	0	48	1	1
8	0	0	0	0	0	0	0	7	42	1
9	0	10	0	3	0	4	0	1	0	32

(d) HMM Test (Speaker 6)										
	0	1	2	3	4	5	6	7	8	9
0	42	0	3	4	1	0	0	0	0	0
1	0	47	0	0	2	0	0	1	0	0
2	24	0	15	10	1	0	0	0	0	0
3	1	0	0	43	0	0	2	0	4	0
4	0	3	1	1	30	7	3	4	1	0
5	0	3	0	0	0	44	0	0	0	3
6	0	0	1	10	6	0	14	3	15	1
7	0	3	2	4	2	0	2	36	0	1
8	0	1	1	3	0	0	0	3	41	1
9	0	13	0	0	0	10	0	4	0	23

6 Inter-Digit Confusion and Insights

From the analysis of the confusion matrices we observe three primary factors that led to model failures during the testing phase while it many pass the validation test.

Rounding Errors beacuse of (VQ): The HMM failed on Digit 2 because it forces sounds into fixed groups. It "rounded" the new speaker's voice so that "Two" sounded like "Zero," causing 24 mistakes. The GMM avoids this by looking at the exact details of the voice, keeping it much more accurate.

High-Frequency Noise: Digit 6 was the hardest for both models because of its high-pitched "hissing" sound. This noise is very similar to the breathy sounds in "Three" and "Eight," making it easy to swap them. This caused many errors because the models could not tell these noisy patterns apart.

simillar ending digits: "Nine" and "One" were often confused because they both end with the same "N" sound. The models focused too much on this identical ending and ignored the different sounds at the start of the words. This caused them to pick the wrong digit by looking only at the end of the word.

Final Conclusion and Insights on the project

Model Suitability

The **Continuous GMM** is clearly the more suited model for this task. It maintains a 97.2% validation score and retains 81% accuracy on an unseen speaker.

Derived Insights

The HMM uses Codebook with 128 buckets which roundoff all into 128 buckets for the accuracy is affected by it as it rounq off the data into 128 buckets. it is not very precise and as the data is also less its unable to perform to full extent . the HMM loses subtle acoustic nuances that distinguish Speaker 6 from the training set.

For isolated digits, the spectral shape (captured by GMM) appears more discriminative than the temporal sequence (captured by HMM)

The drop from 97.2% to 81% (GMM) and 87.2% to 67% (HMM) highlights that Speaker 6 has a different voice that was not fully represented in the 5-speaker training set so the accuracy suffers drastically from validation to test data set accuracies.